PageWork

tutorial 9

⑤

$H(e^{j\omega}) = H_{zp}(\omega)\, e^{-j\omega\alpha}$

a) $(1+z^{-1})^2(1-z^{-1}) = (1+z^{-1}+z^{-2})\cdot(1-z^{-1})$

$= 1+z^{-1}-z^{-2}-z^{-3}$

$H(e^{j\omega}) = 1+e^{-j\omega}-e^{-2j\omega}-e^{-3j\omega}$

$= e^{-3j\omega/2}\left[2j\sin\frac{3\omega}{2} + 2j\sin\frac{\omega}{2}\right]$

$= \underbrace{2\left(\sin\frac{3\omega}{2}+\sin\frac{\omega}{2}\right)}_{H_{zp}(\omega)} e^{j(\pi/2-3\omega/2)}$

$\phi = \pi/2 - 3\omega/2 \quad \omega \in [0, \pi/3]$

b) $(1+z^{-2})(1-z^{-1})^2 = (1+z^{-2})(1-2z^{-1}+z^{-2})$

$= 1-2z^{-1}+2z^{-2}-2z^{-3}+z^{-4}$

$H(e^{j\omega}) = 1-2e^{-j\omega}+2e^{-2j\omega}-2e^{-3j\omega}+e^{-4j\omega}$

$= e^{-2j\omega}(2\cos 2\omega - 4\cos\omega + 2)$

$\phi = -2\omega$

⑥ $h(n) = -h(M-n) \quad M = \text{even}$

$h(n) = 0 \quad \forall\, n < 0$

$h(0) = -h(4) \quad h(1) = -h(3) \quad h(2) = 0$

$h(z) = h(0) + h(1)z^{-1} - h(1)z^{-3} - h(0)z^{-4}$

$h(e^{j\omega}) = e^{-2j\omega}\left[h(0)\cdot 2j\sin 2\omega + h(1)\cdot 2j\sin\omega\right]$

$= e^{j(\pi/2-2\omega)}\left[2h(0)\sin 2\omega + 2h(1)\sin\omega\right]$

$H_{zp}(\omega) = 2h(0)\sin 2\omega + 2h(1)\sin\omega$

$H_{zp}(0.3\pi) = 2h(0)\sin 72 + 2h(1)\sin 54 = 0.2$

$H_{zp}(0.6\pi) = -2h(0)\sin 36 + 2h(1)\sin 72 = 0.8$

$h(0) = -0.165576 \quad h(1) = 0.318253$

⑦ type 3 $M = 6 \quad h(0)\ h(1)\ h(2)$

$[h(0)]^2 + [h(1)]^2 + [h(2)]^2 = 14$

$h(z) = h(0) + h(1)z^{-1} + h(2)z^{-2} + h(2)z^{-4} + h(1)z^{-5} - h(0)z^{-6}$

$h(n) * u(n) = \sum_{k=-\infty}^{\infty} h(n-k)u(k) = \sum_{k=0}^{\infty} h(n-k)$

$= h(n) + h(n-1) + \dots + h(-\infty)$

$y(1) = h(1) + h(0) = 4$

$y(2) = h(1) + h(0) + h(2) = 6 \Rightarrow \boxed{h(2) = 2}$

$h(0) + h(1)\cos\omega + 2\cos 2\omega - 2\cos 4\omega - h(1)\cos 5\omega - h(0)\cos 6\omega$

$= 2h(0)\sin 3\omega \sin 3\omega + 2h(1)\sin 3\omega \sin 2\omega + 2h(2)\sin\omega \sin\omega$

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} 2\sin(3\omega)\,[h(0)\sin 3\omega + h(1)\sin 2\omega + 2\sin\omega]\,e^{j\omega}\,d\omega$$

$= \frac{h[0]}{2} = -1.5 \quad h(0) = 3 \quad h(1) = 1$

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① $f_s = 2f_{max}$

$\text{sinc}\,x = \frac{\sin(\pi x)}{\pi x}$

a) $\text{sinc}(20t) \leftrightarrow \frac{1}{20}\,\text{rect}\left(\frac{\omega}{20}\right) \quad \omega_{max} = 10 \Rightarrow f_s = 20$

b) two doubles when squared $\Rightarrow f_s = 200$

c) $2\pi t = 50\pi \Rightarrow f = 25 \Rightarrow f_s = 50$

d) $f_1 = 15 \quad f_2 = 35 \Rightarrow f_{max} = 35 \Rightarrow f_s = 70$

e) $\text{rect}(300t) \leftrightarrow \frac{1}{300}\sin\left(\frac{f}{300}\right) \Rightarrow f_s = \infty$

② $f = 2000s$

a) No aliasing

b) Aliasing happens

③ ~~$\pi(8000$~~ - sample @ $t = nT = n/F_s$

$$x(n) = \cos\left(\frac{2\pi\,8000\,n}{12000} + \phi\right) = \cos\left(\frac{4\pi}{3}n + \phi\right) = \cos\left(\frac{2\pi n}{3} - \phi\right)$$

$\omega_0 = \frac{2\pi}{3} \quad \theta = -\phi$

④ $f_s = 1000 \quad t = \frac{n}{1000}$

$\phi(n) = \cos\left(\frac{\omega_0 n}{1000}\right) = \cos\left(\frac{\pi}{4}\right) \quad \omega_0 = 250\pi, 2250\pi$

⑤ $\phi(n) = 4\sin(0.4\pi n) - 5\cos(0.48\pi n) + 3\sin(2.4\pi n) + 2\cos(3.52\pi n)$

$3\sin(2.4\pi n) \rightarrow 3\sin(0.4\pi n)$

$2\cos(3.52\pi n) \rightarrow 2\cos(0.48\pi n)$

$= 7\sin(0.4\pi n) - 3\cos(0.48\pi n)$

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⑥ $2\pi f = 160\pi \Rightarrow f = 80 \Leftrightarrow f_s = 160\,Hz$

⑦

⑧ a) $2\Omega_c$

b) $\Omega_c/5$

c) $4\Omega_c$

d) Ω_c

e) Ω_c

⑨ $T = 1/100\ sec$ $\quad 20\pi T = \pi/5 + 2\alpha\pi$

Not unique $\frac{11}{100}$ also works $40\pi T = 2\pi/5 + 2\beta\pi$

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① $X[K] = \sum_{n=0}^{N-1} x[n] e^{-j \cdot \frac{2\pi}{N} n \cdot K}$ N → point DFT

$W_3^K = e^{-j \cdot 2\pi/3 \cdot K}$,

$X[0] = x[0] + x[1] + x[2] = 8$

$X[1] = x[0] + x[1] W_3^1 + x[2] W_3^2 = 1/2 + \sqrt{3}j/2$

$X[2] = x[0] + x[1](-W_3^1) + x[2] W_3^2 = 1/2 - \sqrt{3}j/2$

② $W_4 = e^{-j \cdot 2\pi/4 K} = (-j)^K$

$X[0] = x[0] + x[1] + x[2] + x[3] = 3$

$X[1] = x[0] + x[1] W_4^1 + x[2] W_4^2 + x[3] W_4^3$

$= 3 + 2 + 2j = 5 + 2j$

$X[2] = x[0] + x[1](-1) + x[2] 1 + x[3](-1)$

$= -1$

$X[3] = 5 - 2j$

③ $X[0] = 1$ $X[2] = 1 + 2j$ $X[4] = 2$

$X[1] = 1$ $X[3] = 1$

$X[K] = x^*[N-K]$ → since x is real

$x[5] = x^*[3] = 1$

$x[6] = x^*[2] = 1 - 2j$

$x[7] = x^*[1] = 1$

$\Rightarrow X[K] = 1 + 2j\,\delta[K-2] + \delta[K-4] - 2j\,\delta[K-6]$

$x[n] = \frac{1}{N} \sum_{K=0}^{N-1} X[K] e^{j \cdot \frac{2\pi n}{N} K}$

$= \delta[n] - \frac{1}{2}\sin\left(\frac{\pi}{2} n\right) + \frac{1}{8}(-1)^n \quad 0 \le n \le 7$

⑤ $X[K] = \sum_{n=0}^{N-1} x[n] e^{-j \cdot \frac{2\pi n}{N} K}$

$X[0] = 4$

$X[1] = e^{-j \cdot \frac{2\pi(0)(1)}{4}} + e^{-j(2\pi/4)} + e^{-j(4\pi/4)} + e^{-j(6\pi/4)}$

$= -j + j = 0$

$X[2] = e\,1 - 1 + 1 - 1 = 0$

$X[3] = 0$